

QA Test Report

Due Date Feature * Python Backend

Branch: pod_1_qa | Date: 01 August 2026

Testing Checklist

#	Checklist Item	Result
1	At least one test per acceptance criterion	PASS
2	Edge case - task with no due date	PASS
3	Edge case - task due exactly today	PASS
4	All tests pass against the implementation	FAIL
5	PR description written	PASS

Item 4 fails because the implementation (`models/task.py`, `routes/tasks.py`) has not yet been updated to support `due_date`. The 6 failing tests expose a date format mismatch: the implementation returns a full ISO datetime with timezone (e.g. `2026-08-02T00:00:00+05:30`) while the spec requires a plain `YYYY-MM-DD` string (Assumption A2). This is a bug in the implementation, not in the tests.

Test Coverage by Acceptance Criterion

AC	Description	Tests	Coverage
AC-1	Creating a task with / without a due date	4	PASS
AC-2	Adding a due date to an existing task via PATCH	3	PASS
AC-3	Due date immutability once set (incl. after completion)	5	PASS
AC-4	Sort order - ascending, null tasks last, insertion order preserved	6	PASS
AC-5	Colour indicators (red = overdue)	0	N/A
AC-6	<code>due_date</code> field present on all API responses	3	PASS

AC-5 not tested - colour logic lives entirely in the frontend (`public/app.js`) and is not exercisable via the Flask test client.

Test Run Results (21 tests)

Test Name	AC	Result
<code>test_create_task_with_due_date_returns_201_and_stored_date</code>	AC-1	FAIL
<code>test_create_task_without_due_date_field_defaults_to_null</code>	AC-1	PASS
<code>test_create_task_with_explicit_null_due_date_is_valid</code>	AC-1	PASS
<code>test_create_task_with_due_date_preserves_title_and_completed_defaults</code>	AC-1	FAIL
<code>test_patch_adds_due_date_to_undated_task</code>	AC-2	FAIL
<code>test_patch_due_date_response_contains_all_task_fields</code>	AC-2	FAIL
<code>test_patch_due_date_is_reflected_in_subsequent_get</code>	AC-2	FAIL
<code>test_patch_due_date_rejected_when_already_set</code>	AC-3	PASS
<code>test_patch_due_date_rejected_returns_correct_error_body</code>	AC-3	PASS
<code>test_patch_due_date_rejected_even_when_same_value</code>	AC-3	PASS
<code>test_patch_completed_does_not_modify_due_date</code>	AC-3	FAIL
<code>test_patch_due_date_still_rejected_after_task_is_completed</code>	AC-3	PASS
<code>test_get_tasks_sorted_by_due_date_ascending</code>	AC-4	PASS
<code>test_get_tasks_null_due_dates_sorted_after_all_dated_tasks</code>	AC-4	PASS
<code>test_get_tasks_null_group_preserves_insertion_order</code>	AC-4	PASS
<code>test_get_tasks_overdue_task_appears_before_future_task</code>	AC-4	PASS

test_get_tasks_due_today_sorts_before_future_and_after_overdue	AC-4	PASS
test_get_tasks_mixed_dated_and_undated_full_sort_order	AC-4	PASS
test_create_task_response_always_includes_due_date_field	AC-6	PASS
test_get_all_tasks_response_includes_due_date_on_every_task	AC-6	PASS
test_patch_response_includes_due_date_field	AC-6	PASS

Known Limitations

1. Implementation pending - models/task.py and routes/tasks.py do not yet support due_date. All 6 failures are implementation gaps, not test design issues.
2. Date format bug - implementation returns full ISO datetime with timezone offset instead of plain YYYY-MM-DD string (Assumption A2 violation).
3. AC-5 (colour indicators) has no backend test. Colour logic is frontend-only and cannot be exercised via the Flask test client.
4. Open questions OQ-1 (colours for non-overdue tasks) and OQ-2 (sub-sort within overdue) are unresolved and may require additional tests once decided.